DBMS

ASSIGNMENT-4

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COURSE:- B.TECH CSE (CORE), SECTION-A

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SECTION-A

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Q1. What is transaction in DBMS?

Ans. A transaction is a logical unit of work consisting of multiple database operations (like read, write, update) that are executed together as a single unit to ensure data integrity.

Q2. Explain ACID properties.

Ans. 1) ATOMICITY:- All operations in a transaction are completed or none are.

2) CONSISTENCY:- Database remains valid before and after the transaction.

3) ISOLATION:- Transaction execute independently without interference

4) DURABILITY:- Once committed, changes are permanently stored.

Q3. What is concurrency control?

Ans. Concurrency control is a mechanism that manages simultaneous transactions to prevent conflicts and maintain data consistency.

Q4. What is deadlock?

Ans. A Deadlock occurs when two or more transactions wait indefinitely for each other to release resources, causing a system halt.

Q5. What is serializability?

Ans. Serializability ensures that the result of concurrent transaction execution is equivalent to some serial (one by one) execution.

Q6. What is recovery in DBMS?

Ans. Recovery is the process of restoring the database to a consistent state after a failure using logs and backups mechanisms.

Q7. What is a checkpoint?

Ans. A checkpoint saves the current state of the database to reduce recovery time by limiting how far back the system must go

during recovery.

SECTION-B

Q1. Explain ACID properties using Banking example.

Ans. In a bank transfer from A to B, Atomicity ensures that money is either fully transferred or not at all. Consistency ensures the total balance remains correct before and after the transaction. Isolation ensures that other transactions cannot see intermediate states (like money deducted but not yet added). Durability ensures that once the transaction is completed, the changes are permanently saved even if the system crashes.

Q2. Illustrate a deadlock situation with two transactions:-

Ans. Transaction T1 locks Account A and then tries to lock Account B. At the same time,

Transaction T2 locks Account B and tries to lock Account B. At the same time, transaction T2 locks Account B and tries to lock Account A. Since both transaction are waiting for each other to release locks, neither can proceed, resulting in a deadlock.

Q3. Explain serial vs concurrent execution with example.

Ans.

Serial Execution	Concurrent Execution
1) Transactions execute one after another.	1) Transactions execute simultaneously
2) Speed is slower	2) Speed is faster
3) Easy to maintain data	3) Needs concurrency control.
4) No conflict chances	4) Possible conflicts (like lost update)
5) Simple complexity	5) More complex
eg. T1 completes fully, then T2 starts.	eg. T1 and T2 run at the same time.

SECTION - C

Q1. A Banking system transfers money between accounts. Explain how ACID properties apply.

Ans. During a transfer, Atomicity ensures both debit and credit happens together or not at all. Consistency ensures account balance remain valid (no money loss/ creation). Isolation prevents other transactions from seeing incomplete updates. Durability ensures that once the transaction is successful, it is permanently stored even after system failure.

Q2. Two transaction update same data simultaneously. Identify possible issues :-

Ans. • Lost update :- One transaction overwrites another's update.

• Dirty Read :- A transaction reads uncommitted data from another transaction.

• Non-Repeatable Read :- Same data gives different values when read multiple times.

• Inconsistent Data :- Data integrity is compromised due to improper synchronization.

Q3. A system crashes during transaction. Explain recovery steps:

Ans. • The system checks log files to identify transaction states.

• Undo (Rollback) is applied to incomplete transactions.

• Redo is applied to committed transactions to ensure changes are not lost.

• Database is restored to a consistent state before restarting operations.

SECTION-D

Q1. Explain one ACID property with real life example

Ans Atomicity means "all or nothing".

eg. during an online payment, if money is deducted but the transaction fails, the amount is refunded. Partial transaction is not allowed.

Q2. Draw a transaction timeline and identify conflicts.

Ans. T_1 : Read (x) → Write (x)
T_2 : Read (x) → Write (x)

if T_2 reads or writes x before T_1 completes, a conflict occurs because both transactions access the same data simultaneously, leading to inconsistency.

Q3. Explain one recovery technique

Ans Log based recovery records all transaction operations in a log file. After a crash, the system uses this log to undo incomplete transaction and redo completed one, ensuring the database returns to a consistent state.